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AI-Integrated Digital Twins for IoT-Based Structural Health Monitoring: Emerging Trends and Future Research Directions

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***Abstract*—Artificial Intelligence (AI)-integrated Digital Twins (DTs) are advancing Internet of Things (IoT)-enabled Structural Health Monitoring (SHM) through intelligent sensing, cyber–physical integration, real-time assessment, and predictive infrastructure management. However, the research landscape, thematic evolution, and emerging IoT-driven technologies in AI-enabled DTs for SHM remain insufficiently characterized. A scientometric analysis of 2,930 Scopus-indexed publications, comprising English-language journal articles published between 2018 and 2026 in Computer Science, is presented in the current study. BERTopic, publication and citation analysis, Keyword Co-occurrence Networks (KCN), Country Collaboration Networks (CCN), and Document Co-citation Analysis (DCA) are integrated to examine intellectual structures, collaboration patterns, and technological evolution. Four major research domains are identified: DT Architectures and Intelligent Integration (DTAII), Smart Structural Monitoring and Sensing Systems (SMSS), AI-Enabled Structural Intelligence and Damage Analytics (ASIDA), and Predictive Intelligence and Asset Management (PIAM). Research growth accelerates after 2021, with SMSS emerging as the most mature domain and ASIDA showing rapid expansion driven by deep learning, edge intelligence, and IoT-enabled analytics. Emerging themes include Explainable AI, Physics-Informed Neural Networks, Edge AI, blockchain-enabled DTs, federated DTs, and autonomous predictive maintenance. The findings highlight key research challenges and future directions toward scalable, interoperable, secure, and intelligent IoT-enabled DT frameworks for next-generation SHM.**

***Keywords*—Artificial intelligence, Digital twins, Internet of Things (IoT), Structural health monitoring, Predictive maintenance, Smart infrastructure.**

I. Introduction

Structural Health Monitoring (SHM) is a crucial component for the safety, reliability, and operational performance of critical infrastructure such as bridges, buildings, tunnels, dams, and transportation systems [1]. The growing number of old infrastructure, fast urbanization, environmental pollution, and increasingly frequent natural disasters have raised the need for smart monitoring technologies that can continuously monitor structural conditions [2]. Traditional inspection methods are based on periodic inspections and involve manual labor tasks, which are time-consuming, expensive, and cannot provide real-time information regarding the integrity of the structure [3]. Aging assets, poor maintenance practices, and environmental stresses are becoming a serious problem for critical infrastructure systems around the world. In the United States, over 9% of bridges are structurally deficient [1], and in India, almost 150,000 bridges suffer from maintenance backlogs [4]. In the US, the deterioration of water infrastructure causes nearly 6.9 billion hours of traffic delay each year and losses in water utility networks of 30-50% of treated water through leakage and operational inefficiencies [2]. Furthermore, operational and maintenance-related issues account for approximately 58% of small dam failures worldwide [3], while land subsidence has emerged as a growing threat to urban infrastructure, adversely affecting building foundations, transportation networks, and utility systems. For such reasons, advanced continuous monitoring and predictive maintenance technologies are becoming an important research priority. A potential solution to such challenges is Digital Twin (DT) technology [5]. A DT is a dynamic virtual copy of a physical asset that can continuously exchange information with the real asset using technologies such as sensing, communication, and computation [6]. DTs leverage real-time data from sensors connected to the infrastructure asset and combine them with a simulation and analytical model to continuously monitor and assess the condition, diagnose faults, and predict performance throughout the asset's life cycle [7]. DTs' ability to offer real-time visibility of structural behaviour has helped to speed their adoption in smart cities, transportation systems, manufacturing, healthcare, and civil infrastructure management [8]. With the rapid development of the Internet of Things (IoT), Wireless Sensor Networks (WSN), Cloud Computing, Edge Computing, and Cyber-Physical Systems, the DT's capabilities have been further strengthened [9]. Embedded sensors, unmanned aerial vehicles (UAVs), imaging sensors, and monitoring platforms generate a large amount of heterogeneous data from modern infrastructure systems. However, the challenge lies in fitting these large amounts of data into a coherent structural intelligence that is actionable [10]. To overcome it, DT frameworks are increasingly combined with Artificial Intelligence (AI) techniques such as machine learning, Deep Learning (DL), computer vision, reinforcement learning, and physics-informed networks. AI-driven DT systems can automatically

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[1] https://infrastructurereportcard.org/

[2] https://sevenseaswater.com/

[3] https://ui.adsabs.harvard.edu

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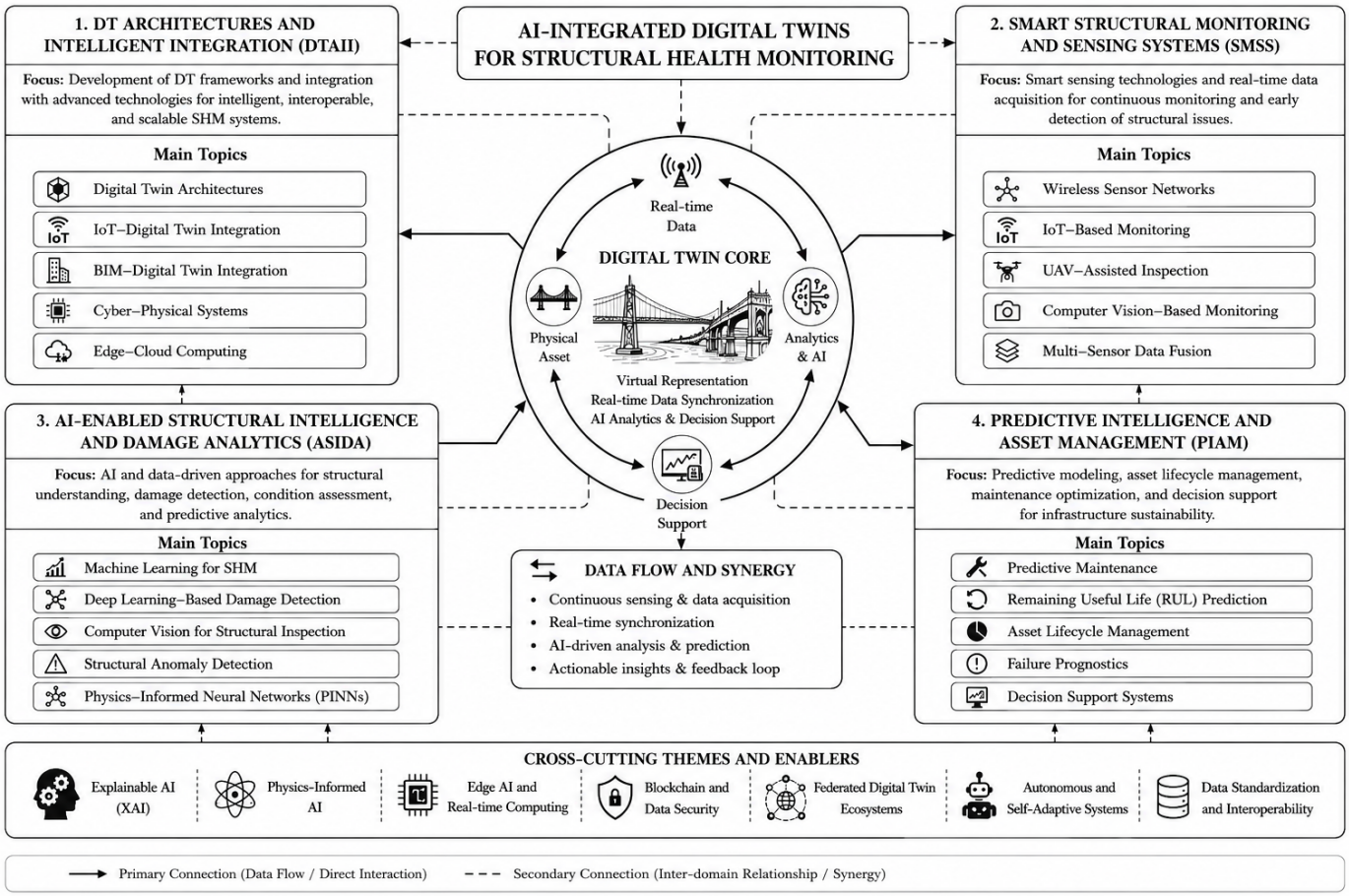


Fig. 1: Conceptual overview of the major research domains and active research areas in AI-integrated DTs for SHM.

identify structural anomalies and damage patterns and provide estimates of structural conditions, enabling more resilient and intelligent infrastructure management [11]. The AI, IoT, and DT technologies have greatly sped up the evolution of Intelligent SHM systems [12]. With the recent developments of DT architecture, smart sensing techniques, damage analysis, predictive maintenance, and asset management, the field of SHM has become a highly interdisciplinary research area [13]. As research in the field has progressed rapidly, however, it has become more challenging to gain a comprehensive understanding of the evolution of the field, key research themes, and areas of new growth. While several review studies have focused on DT technologies, AI applications, and SHM methodologies separately, a complete scientometric view of AI-integrated DT technologies for SHM is still limited. Therefore, the present study aims to provide a systematic overview of the research landscape of AI-integrated DTs for SHM. The literature is presented in a structured manner under the four major research domains: DT Architectures and Intelligent Integration (DTAII), Smart Structural Monitoring and Sensing Systems (SMSS), AI-Enabled Structural Intelligence and Damage Analytics (ASIDA), and Predictive Intelligence and Asset Management (PIAM). These domains are the basic concepts that enable the creation of smart, resilient, and data-driven SHM systems. Figure 1 represents the conceptual overview of AI-integrated DTs for SHM. It represents the interconnection of the four major research domains and their active research areas, the DT core, and the enabling technologies of intelligent monitoring, damage assessment, and predictive asset management.

A. Digital Twin Architectures and Intelligent Integration

DTAII are the core components of the AI-Based SHM systems. A DT is a virtual model of a physical system that is updated in real-time with data from sensors, communication systems, and computational models [14]. The architecture of the DT is a key factor in its effectiveness, impacting the collection of data, the sharing of information, system interoperability, and decision-making in real-time. The modern DT architecture combines several technologies such as the IoT, Building Information Modeling (BIM), cyber-physical systems (CPS), and cloud-edge computing to create seamless connectivity between physical and virtual environments [15]. Intelligent integration has greatly improved the capabilities

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TABLE I: Evolution of DTAII Research

Year	Research Focus	Challenges	Adoption
2026	Federated DTs and autonomous DT ecosystems	Interoperability and trust management	High
2025	Blockchain-enabled DTs and intelligent integration	Cybersecurity and privacy concerns	High
2024	Edge-cloud DT architectures and real-time synchronization	Scalability and latency issues	High
2023	BIM-DT integration and smart infrastructure modeling	Data interoperability and standardization	High
2022	CPS and IoT-DT integration	Heterogeneous sensor data management	Moderate
2021	Cloud-based DT platforms	Computational complexity and data processing	Moderate
2020	Digital Twin reference architectures	Lack of standardized frameworks	Moderate
2019	Virtual infrastructure modeling	Limited real-time connectivity	Low
2018	Early Digital Twin frameworks	Adoption barriers and technology maturity	Low

TABLE II: Evolution of SMSS Research

Year	Research Focus	Challenges	Adoption
2026	Edge AI monitoring and intelligent sensor ecosystems	Real-time processing of large-scale data	High
2025	Autonomous monitoring systems and digital sensing platforms	Communication reliability	High
2024	Multi-sensor fusion and UAV-assisted inspection	Sensor synchronization and data quality	High
2023	Computer vision-based monitoring	Environmental noise and occlusion	High
2022	IoT-based structural monitoring	Data transmission and scalability issues	Moderate
2021	Wireless sensor networks	Energy consumption and battery limitations	Moderate
2020	Smart sensor technologies	Sensor reliability and calibration	Moderate
2019	Remote structural monitoring	Limited communication infrastructure	Low
2018	Sensor-based SHM systems	Restricted monitoring coverage	Low

of DT systems in recent years. The continuous monitoring of structural conditions is possible through an IoT platform with sensing equipment, and BIM allows for a detailed digital representation of infrastructure assets during their life cycles [16]. Additionally, CPS frameworks facilitate bidirectional information exchange between the physical and digital worlds, facilitating real-time updates and adaptive responses [14]. The arrival of edge-cloud computing has further enhanced scalability, computational efficiency, and data processing capabilities, enabling DT systems to handle large-scale infrastructure monitoring applications [17]. Therefore, DTAII is an important research area to realize intelligent, interoperable, data-driven SHM solutions. The evolution of the research in the field of DTAII is highlighted in Table I for the period 2018-2026 [14][18][19]. The initial research was on a basic DT framework and Virtual Infrastructure modeling. The study grew to be more focused on the integration of IoT-DT, cyber-physical systems, and cloud-based platforms to facilitate connectivity and data exchange. Recent advancements focus on BIM integration, edge-cloud computing, blockchain-based DTs, and federated DT ecosystems for SHM applications to improve interoperability, scalability, and smarter decision-making.

B. Smart Structural Monitoring and Sensing Systems

Real-time data is the building block for AI-supported DT frameworks, where SMSS serve as the data foundation. It allows for the continuous monitoring of how the structure is reacting to its environment, which can aid in early detection of anomalies, performance evaluation, and maintenance decisions [20]. In the modern SHM system, WSNs, monitoring devices with IoT technology, UAV, computer vision technique, and multi-sensor fusion are adopted to collect various structural and environmental information [12]. The technologies help to enhance the precision of monitoring, spatial coverage, data reliability, and decrease manual effort for monitoring. With the development of edge computing and intelligent sensing platforms, the capability of SMSS has been further improved, enabling low-latency processing of data and ongoing synchronisation with DTs [21]. The interconnection of different sensing technologies can provide a complete picture of complex infrastructure such as bridges, tunnels, buildings, and transportation systems, which increases situational awareness and infrastructure resilience [22]. The evolution of research in SMSS (as shown in Table II [23][24]) shows that it has shifted from traditional sensor-based SHM systems to intelligent sensing ecosystems. Previous research has focused mainly on the basic deployment of sensors and remote monitoring, while current research has focused on monitoring via IoT, computer vision, UAV-assisted inspection, multi-sensor fusion, autonomous monitoring systems, and Edge AI for real-time structural intelligence. Such a change mirrors the maturing and progress of smart sensing technologies in next-generation SHM systems.

C. AI-Enabled Structural Intelligence and Damage Analytics

ASIDA turns raw monitoring data into actionable insights for structural condition assessment and decision-making. By integrating AI with DT technology, ASIDA enables automated damage detection, structural anomaly identification, condition assessment, and health prediction [25]. The adoption of machine learning, DL, computer vision, and Physics-Informed Neural Networks (PINNs) has emerged as a crucial technology in handling intricate structural information to boost predictive accuracy and facilitate intelligent infrastructure management [26]. Recent advances in AI have significantly enhanced the capability of DT-based SHM systems by enabling continuous learning from heterogeneous sensor data and visual inspections [27]. These can help in the timely detection of damage, reduce manual inspection needs, and increase the accuracy of predictive analytics of critical infrastructure. As a result, AI-based structural intelligence is becoming an integral element of the next-generation smart infrastructure systems.

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TABLE III: Evolution of ASIDA Research

Year	Research Focus	Challenges	Adoption
2026	Explainable AI and Foundation Models for SHM	Trustworthiness and interpretability	High
2025	Physics-Informed Neural Networks (PINNs)	Model complexity and training cost	High
2024	Deep learning-based damage detection	Computational requirements	High
2023	Computer vision for structural inspection	Image variability and environmental effects	High
2022	Structural anomaly detection	False positives and uncertainty	Moderate
2021	Transfer learning for SHM	Limited labeled datasets	Moderate
2020	Machine learning-based damage assessment	Feature engineering challenges	Moderate
2019	Data-driven structural diagnostics	Data imbalance issues	Low
2018	AI-assisted structural evaluation	Limited data availability	Low

Table III shows advancement in research areas within ASIDA from traditional AI-driven structural assessment to explainable and physics-informed intelligence [28][29][30]. Early research mainly concentrated on machine learning for damage assessment and structural diagnostics, while recent research has focused on DL, computer vision, structural anomaly detection, Explainable AI (XAI), PINNs, and enhancing the accuracy of predictions and model explainability and trustworthiness in SHM.

D. Predictive Intelligence and Asset Management

PIAM extends DT-enabled SHM from condition assessment to proactive maintenance and lifecycle management. By integrating AI, predictive analytics, and DT technologies, PIAM enables continuous assessment of infrastructure health, prediction of structural degradation, and optimization of maintenance strategies [31]. The primary focus of research is on predictive maintenance, asset lifecycle management, failure prognostics, and intelligent decision support systems that enhance the reliability of infrastructure, its operational efficiency, and sustainability [32]. In recent years, the integration of historical data, real-time sensor readings, and AI-driven predictive models has made it possible for DTs to aid in data-driven maintenance planning [33]. Such capabilities support early detection of potential failure, efficient resource utilization, lower maintenance expenses, and increased critical infrastructure resilience during operations. In terms of research evolution, the transition of research from traditional maintenance scheduling and condition-based maintenance to intelligent predictive maintenance and autonomous asset management can be seen from the research described in Table IV [34][35][36]. The integration of predictive intelligence into sustainable infrastructure management is gaining momentum, with recent studies increasingly addressing topics such as Remaining Useful Life (RUL) prediction, infrastructure resilience assessment, maintenance optimization for risk awareness, and AI-aided decision support systems.

TABLE IV: Evolution of PIAM Research

Year	Research Focus	Challenges	Adoption
2026	Autonomous asset management systems	Decision-making under uncertainty	High
2025	Risk-aware maintenance optimization	Multi-objective optimization complexity	High
2024	Infrastructure resilience assessment	Accurate resilience quantification	High
2023	Remaining useful life prediction	Dynamic operational conditions	High
2022	Predictive maintenance frameworks	Forecasting accuracy	Moderate
2021	Failure prognostics	Degradation modeling challenges	Moderate
2020	Asset lifecycle management	Data integration and management issues	Moderate
2019	Condition-based maintenance	Limited historical data availability	Low
2018	Maintenance scheduling systems	Reactive maintenance practices	Low

E. Research Questions

The current research conducts a comprehensive scientometric analysis of the AI-integrated DT research in the field of SHM in the Scopus database from 2018 to 2026. The goal of the study is to systematically identify the intellectual structure, the evolution of research, the collaboration patterns, and emerging trends of the fast-growing interdisciplinary area. To achieve such a goal, the following research questions are formulated:

1) **RQ1:** What are the major research domains and thematic areas in AI-integrated DT-enabled SHM?
2) **RQ2:** Which are the most active research topics in each of the four identified domains: DTAII, SMSS, ASIDA, and PIAM?
3) **RQ3:** How has the AI-integrated DT research landscape evolved in terms of publication growth, citation impact, geographical distribution, leading journals, institutions, and countries during 2018-2026?
4) **RQ4:** What are the intellectual foundations, collaboration patterns, and thematic structures of the field as derived from Keyword Co-occurrence Networks (KCN), Country Collaboration Networks (CCN), and Document Co-citation Analysis (DCA)?
5) **RQ5:** What are the emerging research trends, existing challenges, and future research opportunities for AI-integrated DT technologies in SHM?

F. Research Contributions

The current study utilizes scientometric methodologies to systematically reveal the intellectual structure, evolution of research, collaboration landscape, and emerging trends in the field, unlike traditional narrative reviews that mainly summarize methods and applications. The major contributions of the current study are summarized as follows:

1) An extensive scientometric methodology is established to systematically trace the development of DT research

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that integrates with AI for SHM based on the Scopus database.
2) Four distinct research areas are explored and examined, namely DTAII, SMSS, ASIDA, and PIAM, which gives a structured understanding of the field of AI-integrated DT in SHM.
3) A conceptual overview is proposed to illustrate the interconnections among the four identified research domains, their major active research areas, enabling technologies, and future research directions.
4) The scientific field of AI-assisted DT development is systematically explored using the analysis of publication growth, citation patterns, prominent journals, productive countries, and scientific geographic distribution.
5) The intellectual and collaborative structure of the research field is explored using multiple scientometric techniques, including KCN, CCN, DCA, and cluster analysis to identify influential research themes and emerging trends.
6) The study summarizes current challenges, identifies promising future research directions, and provides a comprehensive roadmap for researchers and practitioners toward the development of intelligent, explainable, resilient, and sustainable DT-enabled SHM.

Overall, the current study provides a comprehensive scientometric overview of AI-integrated DT research for SHM, revealing its intellectual structure, research trends, collaboration patterns, and emerging themes. The results provide important insights into future studies and the design of smart, withstanding, and sustainable SHM systems.

a) Paper Organization:: The rest of the current study is organized as follows. Section 2 reviews the existing literature on AI-integrated DT for SHM. Section 3 describes the research methodology, with the data collection process and scientometric techniques followed. The scientometric results are presented in Section 4, which includes publication trends, citation analysis, CCN, KCN, DCA, and thematic analyses. The main findings, research limits, and future research directions are discussed in Section 5. Finally, the paper is concluded in Section 6.

II. Literature Review

Although numerous studies have reviewed DT, AI, and SHM, most focus on specific technologies, application domains, or case studies. Fully integrated AI-driven DT and SHM studies are still scarce. Table V summarizes representative studies published in high-impact journals and identifies the databases used, methodologies employed, major research gaps, and key contributions. The comparison indicates that the literature reviews and application-oriented frameworks have been the main focus of previous research, with a lack of attention to comprehensive research overviews, thematic development, collaboration structures, and research trends. These gaps are the motivation for the present scientometric study.

III. Research Methodology

The overall research methodology adopted in the current study is illustrated in Figure 2. The research methodology comprises four stages: research formulation, data collection, data analysis, and interpretation of results. The process combines scientometric analysis with BERTopic-based topic modeling to systematically analyze the research landscape of AI-integrated DTs for SHM.

1) **Research formulation:** The first stage establishes the scope and objectives of the study by identifying research gaps in the existing literature on AI-integrated DTs for SHM. Based on the gaps, the research questions and objectives were formulated to investigate the evolution, knowledge structure, collaboration patterns, and emerging research directions of the field.
2) **Data acquisition:** The literature dataset was retrieved from the Scopus database using a comprehensive search query containing keywords related to DTs, AI, SHM, IoT, Machine Learning, DL, Damage Detection, and Predictive Maintenance. The research articles considered were only those published in English between the years 2018 and 2026 and belonging to the Computer Science subject area. The retrieved bibliographic records were imported into CiteSpace. Data preprocessing was applied, in which duplication of data was carried out to ensure consistency of the data. There were no matching records found in the data set retrieved. A total of 2,930 publications were selected after applying the predefined inclusion and exclusion criteria. The bibliographic records, such as title, abstract, and author keywords, were then analysed using BERTopic to extract the most prominent research themes. Four main research areas were identified by BERTopic: DTAII, SMSS, ASIDA, and PIAM, as shown in Figure 3. These domains were used to create category-specific searches in Scopus that produced 641, 1010, 883, and 393 publications for in-depth scientometric analysis for DTAII, SMSS, ASIDA, and PIAM, respectively.
3) **Data analysis:** Descriptive scientometric analysis and thematic content analysis were used to analyze the collected publications. The evolution and intellectual structure of the research field were examined by using publication trends, citation patterns, KCN, CCN, and DCA.
4) **Results and interpretation:** Finally, the findings obtained from the scientometric and thematic analyses were synthesized to provide a comprehensive overview of AI-integrated DT research for Structural Health Monitoring. The results reveal publication trends, influential studies, collaboration networks, research hotspots, and future research opportunities across the four identified research domains.

A. Computational Metrics

To quantitatively evaluate the research landscape of AI-integrated DTs for SHM, three scientometric analyses were performed: KCN, CCN, and DCA. The computational metrics employed in each analysis are described as follows.

1) **Keyword Co-occurrence Network**

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TABLE V: Representative studies on AI-integrated Digital Twins for Structural Health Monitoring.

Refs.	Database	Method	Major Gaps	Prominent Solutions
Fuller et al. [37]	Scopus	Narrative Review	Limited SHM and infrastructure applications.	Established DT concepts and enabling technologies.
VanDerHorn and Mahadevan [38]	Scopus, WoS	Comprehensive Literature Review	Limited infrastructure-specific applications.	Standardized DT definitions and lifecycle framework.
Thelen et al. [39] (Part I)	Scopus	Comprehensive Literature Review	Limited SHM-specific discussion.	Unified DT modeling and enabling technologies.
Thelen et al. [40] (Part II)	Scopus	Comprehensive Literature Review	Limited civil infrastructure applications.	Reviewed uncertainty quantification and DT optimization.
Omrany et al. [41]	Scopus	Systematic Literature Review	Construction-oriented; limited SHM focus.	Synthesized DT adoption strategies in construction.
Karabulut et al. [42]	Scopus	Systematic Literature Review	Focused only on ontology-driven DTs.	Introduced ontology-based semantic DT integration.
Chang et al. [5]	Scopus	Systematic Literature Review	Transportation-focused; limited SHM applications.	Summarized DT technologies for transportation infrastructure.
Sun et al. [43]	Scopus, WoS	Comprehensive Literature Review	No scientometric or thematic analysis.	Advanced DT architectures for intelligent SHM.
Mardanshahi et al. [10]	Scopus	State-of-the-Art Review	Focused mainly on sensing technologies.	Evaluated next-generation sensing technologies for SHM.
Qiu et al. [44]	Scopus	Comprehensive Literature Review	Limited AI-driven structural analytics.	Promoted sustainable DT-enabled infrastructure management.
Wang et al. [45]	Scopus	PRISMA-based Systematic Review	Limited SHM-specific thematic mapping.	Developed a DT implementation roadmap for construction.
Tao et al. [46]	Scopus, WoS	Comprehensive Literature Review	Manufacturing-oriented; limited SHM applications.	Developed a DT-driven intelligent manufacturing framework.
Lu et al. [36]	Scopus, WoS	Comprehensive Literature Review	Limited SHM and infrastructure applications.	Proposed a DT reference architecture and implementation model.

Keyword co-occurrence analysis identifies the relationships among frequently occurring keywords and reveals the major research themes.

- **Occurrence Frequency (OF):**
  The occurrence frequency of a keyword is calculated as

$$OF(k_i) = \sum_{d=1}^{N} I(k_i, d), \tag{1}$$

  where N denotes the total number of publications and

$$I(k_i, d) = \begin{cases} 1, & \text{if keyword } k_i \text{ appears in document } d, \\ 0, & \text{otherwise.} \end{cases}$$

- **Total Link Strength (TLS):**
  The total link strength of a keyword is defined as

$$TLS(k_i) = \sum_{j=1, j\neq i}^{m} L(k_i, k_j), \tag{2}$$

  where m is the total number of keywords and $L(k_i, k_j)$ represents the co-occurrence frequency between keywords k_i and k_j.
- **Betweenness Centrality (BC):**
  The importance of a keyword within the network is measured by

$$BC(v) = \sum_{s\neq v\neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}, \tag{3}$$

  where σ_{st} is the total number of shortest paths between nodes s and t, while $\sigma_{st}(v)$ denotes the number of those paths passing through node v.

2) **Country Collaboration Network**
Country collaboration analysis evaluates the global research collaboration among countries.

- **Number of Publications (NP):**
  The publication productivity of a country is computed as

$$NP(c) = \sum_{d=1}^{N} I(c, d), \tag{4}$$

  where

$$I(c, d) = \begin{cases} 1, & \text{if country } c \text{ contributed to publication } d, \\ 0, & \text{otherwise.} \end{cases}$$

- **Collaboration Frequency (CF):**
  The collaboration frequency between the two countries is given by

$$CF(c_i, c_j) = N_{ij}, \tag{5}$$

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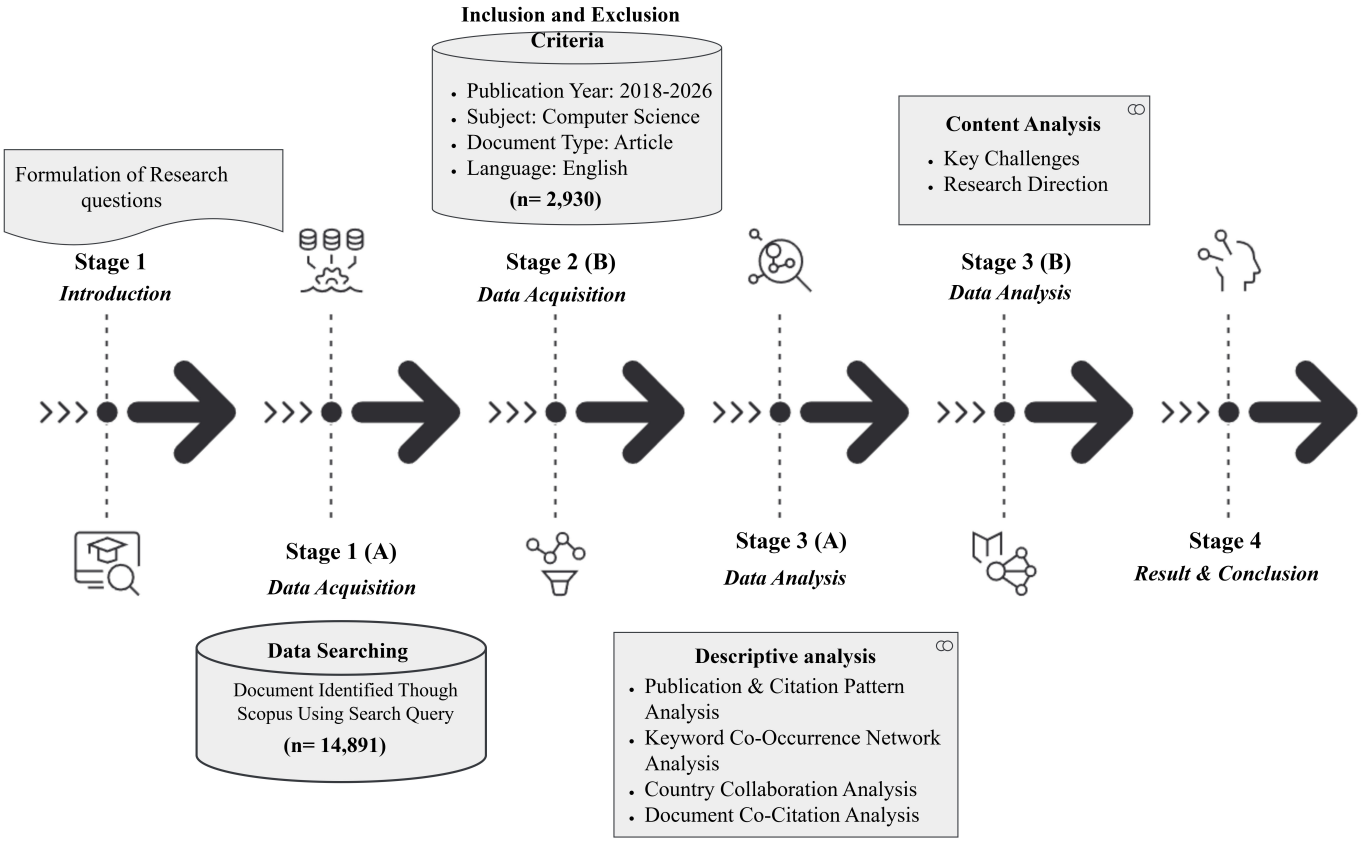


Fig. 2: Research Methodology.

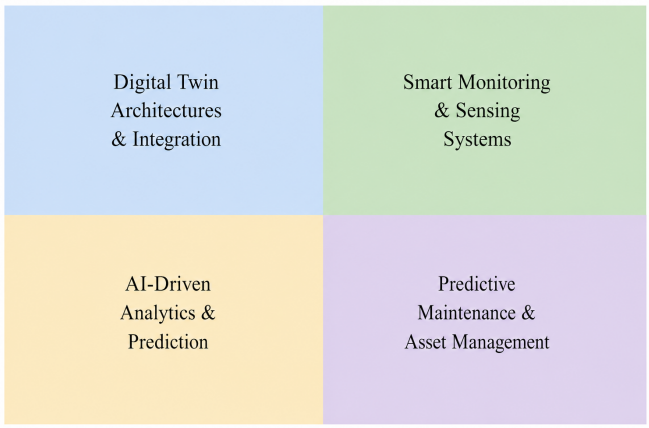


Fig. 3: Major research themes identified using the BERTopic framework.

where N_{ij} denotes the number of jointly authored publications between countries c_i and c_j.

- **Total Link Strength (TLS):**
  The collaboration strength of a country is computed as

$$TLS(c_i) = \sum_{j \neq i} N_{ij}. \tag{6}$$

- **Betweenness Centrality (BC):**
  The importance of a country within the collaboration network is calculated as

$$BC(c) = \sum_{s \neq c \neq t} \frac{\sigma_{st}(c)}{\sigma_{st}}. \tag{7}$$

3) **Document Co-citation Analysis**
Document co-citation analysis identifies influential publications and reveals the intellectual structure of the research field.

- **Co-citation Frequency (CF):**
  The co-citation frequency between two documents is defined as

$$CF(d_i, d_j) = N_{ij}, \tag{8}$$

where N_{ij} is the number of publications citing both documents simultaneously.

- **Citation Count (TC):**
  The citation count of a document is calculated as

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$$TC(d_i) = \sum_{k=1}^{N} I(d_i, k), \tag{9}$$

where

$$I(d_i, k) = \begin{cases} 1, & \text{if publication } k \text{ cites document } d_i, \\ 0, & \text{otherwise.} \end{cases}$$

- **Betweenness Centrality (BC):**
The bridging importance of a publication is computed as

$$BC(d) = \sum_{s \neq d \neq t} \frac{\sigma_{st}(d)}{\sigma_{st}}. \tag{10}$$

- **Modularity (Q):**
The modularity index evaluates the quality of network clustering and is defined as

$$Q = \frac{1}{2m} \sum_{i,j} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta(c_i, c_j), \tag{11}$$

where A_{ij} is the adjacency matrix, k_i and k_j denote the node degrees, m is the total number of edges, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same cluster; otherwise, it is 0.
- **Silhouette Score (S):**
The cluster consistency is measured using.

$$S(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \tag{12}$$

where $a(i)$ is the average intra-cluster distance and $b(i)$ is the minimum average inter-cluster distance. A silhouette value close to 1 indicates highly cohesive and well-separated clusters.

B. Data Source and Visualization Tool

The most widely used bibliographic databases for scientometric research include Scopus, Web of Science (WoS), and Google Scholar [47]. Scopus was selected as the primary information source because of its broad coverage of peer-reviewed publications, comprehensive bibliographic metadata, relatively rapid indexing, and advanced search capabilities [48]. Moreover, Scopus offers several flexible filters, such as year, subject area, document type, country, institution, and author, and can be used for large-scale scientometric studies [49]. Scopus offers wider coverage of disciplines compared to other major databases such as WoS and IEEE Xplore, and structured bibliographic metadata allowing consistent analysis of the analysis across different research areas [49]. In addition, the researchers had an institutional subscription to the Scopus database that allowed them to retrieve and analyse the required literature systematically through the bibliographic records. The retrieved bibliographic data were then pre-processed and analyzed in Microsoft Excel, VOSviewer, and CiteSpace [50]. The data was preprocessed in Microsoft Excel for descriptive statistical analysis, and CiteSpace was used to perform DCA, CCN, and bibliographic data preprocessing (duplicate detection and network construction), and KCN was constructed and visualized in VOSviewer.

IV. Data Analyses

The current section presents the scientometric analysis of the selected research domains to identify research trends, influential contributions, collaboration patterns, and emerging research directions.

A. Publication Pattern Analysis

Publication pattern analysis provides an overview of the growth and maturity of a research domain by examining annual publication trends. Figure 4 and Table VI present the publication growth for DTAII, SMSS, ASIDA, and PIAM during 2018-2026. In Table VI, t denotes the number of publications in a particular year, t% represents the percentage contribution of that year to the total publications within the corresponding research area, TP indicates the total number of publications, and TP% denotes the percentage contribution of each research area to the overall dataset. The results show a consistent increase in publication output across the four research areas, indicating the growing research interest in AI-integrated DT technologies. The overall publishing activity in 2018-2020 was quite low and then sharply rose after 2021 due to the quick growth in AI, IoT, intelligent sensing, and DT technologies. In terms of publications, SMSS is the most active research domain, having a total of 1010 publications (34.47%), followed by ASIDA with 883 publications (30.17%), DTAII with 641 publications (21.89%), and PIAM with 393 publications (13.41%). The publication trend shows that the number of publications per year for SMSS was the highest throughout the study period, and the number of publications per year for ASIDA showed the highest growth after 2022, which is considered an indication of the growing use of AI-based structural diagnostics and damage assessment techniques. DTAII also saw significant growth, especially since 2022, evidencing the rise in significance of DT architectures and intelligent system integration. PIAM had the smallest percentage of publications, but showed a steady increase in publications, suggesting an increasing focus on predictive maintenance and infrastructure asset management. The highest number of publications was generated in each of the four research areas in 2025 before it slightly decreased in 2026. The reduction is not a reflection of a drop in research interest, as it reflects only a partial indexing of the publication year 2026 when data was collected. Overall, the publication trends show that the focus of research has slowly progressed from basic DT architectures to intelligent sensing, AI-powered analytics, and predictive infrastructure management. The high proportion of publications from SMSS and ASIDA (around 64.64%) underscores their important contributions to the field of AI-driven DT.

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TABLE VI: Growth in each research area

Research Area	2018		2019		2020		2021		2022		2023		2024		2025		2026		Overall	
	t	$t\%$	t	$t\%$	t	$t\%$	t	$t\%$	t	$t\%$	t	$t\%$	t	$t\%$	t	$t\%$	TP	TP%		
DTAII	1	0.16	2	0.32	8	1.23	15	2.34	39	6.08	82	12.80	92	14.35	213	33.23	189	29.49	641	21.89
SMSS	6	0.59	10	0.99	33	3.27	58	5.74	91	9.01	135	13.37	173	17.13	284	28.12	220	21.78	1010	34.47
ASID	3	0.34	2	0.23	17	1.93	33	3.74	70	7.93	118	13.36	143	16.19	269	30.46	228	25.82	883	30.17
PIAM	2	0.51	4	1.02	12	3.05	19	4.83	27	6.87	44	11.20	60	15.27	125	31.81	100	25.45	393	13.41

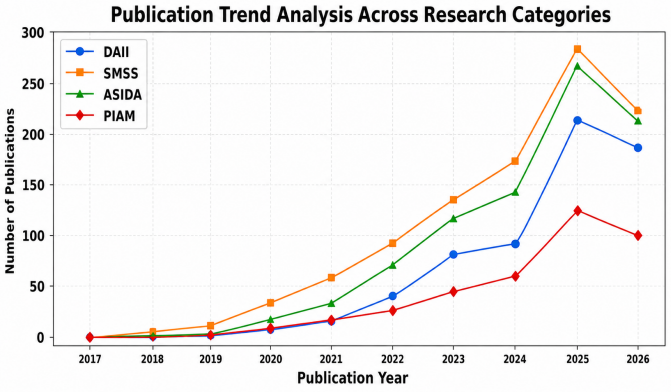


Fig. 4: Publication growth for DTAII, SMSS, ASID, PIAM.

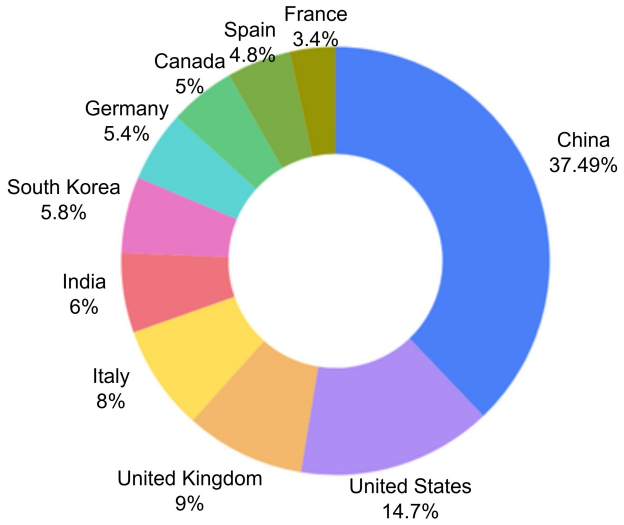


Fig. 5: Geographical Distribution of Publications

1) Global Distribution: Global distribution analysis describes how countries contribute to the selected subject areas. Overall distribution of publications is shown in Figure 5, and the top five countries by number of publications in each area of research with their percentage contribution are shown in Table VII. Overall distribution reveals that China is the top country accounting for 37.49% of the total publications, followed by the United States with 14.7%, the United Kingdom with 9%, Italy with 8%, India with 6%, South Korea with 5.8%, Germany with 5.4%, Canada with 5%, and Spain with 4.8%, which is followed by France at 3.4%. As illustrated in Table VII, China is the first country in each research area, with its share in DTAII 13.72%, in SMSS 18.54%, in ASIDA 18.94%, and in PIAM 15.66%, respectively, indicating that China is the leader in AI-integrated DT research. In other categories such as SMSS, ASIDA, the United States is the second largest contributor, while the United Kingdom is the second largest in DTAII and PIAM. Italy has a solid and stable engagement in each area, and Germany, Spain, and India play a crucial role in certain areas. The geographical distribution shows that research is largely focused upon technologically advanced countries that have significant investment in technologies such as AI, DT, and Smart Infrastructure. The total number of publications in China, the United States, and the United Kingdom is the largest and most significant contributor, playing a vital role in the development of AI-based SHM research.

B. Citation Pattern Analysis

Citation analysis is a way to measure the influence and impact of publications in each research field. Table VIII summarizes the citation performance metrics, which include TP (total publications), TC (total citations), CPP (citations per publication), i-10 index, and the h-index. SMSS is the most influential and mature research domain among the four research areas with 1,010 publications, 28,689 citations, 28.40 CPP, i-10 index 454, and h-index 85. ASIDA is the second most cited in 883 publications and 21,389 citations of 24.22 and an h-index of 72, which indicates the increasing relevance of structural intelligence and damage analytics by means of AI. Even though DTAII has several publications (641), it has the lowest CPP (16.50) and h-index (52), indicating that this field is comparatively emerging even with rising publications. Despite having the lowest number of publications, there are 9,701 citations cited by the publications, and the average citation per publication is 24.68, making the impact of each publication more significant than that of DTAII, while there are 55 publications with an h-index of 55. Overall, the citations analysis indicates that the most influential scientific studies are with SMSS, with ASIDA and PIAM also having high research impact. The relatively few citation scores of DTAII indicate that there are significant research opportunities, as well as greater citations needed.

1) Prestigious Journal: Prominent journals have been analyzed to determine the key journals that are contributing to the research on AI-integrated DT. Table IX lists the top journals in each research area, according to the number of citations (c), the number of publications (np), the average number of citations per publication (ACPP), the share of papers published

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TABLE VII: Research area-wise top productive nations

Research Area	Country 1	Country 2	Country 3	Country 4	Country 5
DTAII	China (135, 13.72%)	United Kingdom (87, 8.84%)	United States (81, 8.23%)	Italy (57, 5.79%)	Spain (53, 5.39%)
SMSS	China (272, 18.54%)	United States (125, 8.52%)	Italy (92, 6.27%)	United Kingdom (90, 6.13%)	Germany (63, 4.29%)
ASID	China (257, 18.94%)	United States (145, 10.68%)	United Kingdom (109, 8.03%)	Italy (61, 4.50%)	India (55, 4.05%)
PIAM	China (86, 15.66%)	United Kingdom (42, 7.65%)	Italy (40, 7.29%)	United States (34, 6.19%)	Germany (27, 4.92%)

TABLE VIII: Citation analysis of each research area

Research Area	**TP**	**TC**	**CPP**	**i-10 Index**	**h-index**
DTAII	641	10582	16.50	217	52
SMSS	1010	28689	28.40	454	85
ASID	883	21389	24.22	30	72
PIAM	393	9701	24.68	162	55

in each journal (np%), and the impact factor (IF). The articles published by Applied Sciences Switzerland and IEEE Access were the most, with 47 and 44, respectively, while Advanced Engineering Informatics had the highest impact factor of 9.9. Applied Sciences Switzerland has the most publications (82) and citations (4,185) among SMSS, followed by IEEE Access (72) and Sensors (68), while IEEE Internet of Things Journal has the highest impact factor (8.9) among the journals. The citation counts for ASIDA, IEEE Access, Applied Sciences Switzerland, and other journals were compared, and IEEE Access was ranked first with 2,129 citations and 54 publications, followed by Applied Sciences Switzerland with 53 publications. Again in 2018, Applied Sciences Switzerland ranks as the top journal in terms of publications for PIAM, with 40, followed by IEEE Access (30) and Sensors (20). Applied Sciences Switzerland is consistently among the journals with the most productivity in all areas of research, as well as IEEE Access, Sensors, and Sustainability Switzerland, and the IEEE Internet of Things Journal, which are high-impact venues for research on the DT integrated with AI.

C. Keyword Co-occurrence Network Analysis

The KCN analysis is a popular scientometric method for revealing intellectual structure, key research themes, and emerging trends in a research field. It looks at how frequently keywords are co-occurring in publications and therefore shows the correlations between different research topics. VOSviewer (Version 1.6.20) was used to create the KCNs in the present study, with the node size based on the frequency of the keywords, the links showing co-occurrence relationships, and the strength of the links indicating the level of association between the keywords. In the diagram, keywords that have larger clusters and closer connections are well-established research themes, while smaller clusters are emerging topics.

1) Keyword Co-occurrence Network Analysis for DTAII : The KCN analysis for DTAII is shown in Figure 6, and the most influential keywords and thematic clusters are presented in Tables X and XI, respectively. The most frequently used term and the term with the most links is DT, which has the highest frequency (393) and the highest number of links (1939). It is the central term in the network and links various research themes. Table X shows that the IoT (frequency 95, link strength 642) and Decision Making (frequency 90, link strength 594) demonstrate strong relationships with DT, highlighting the need for real-time data collection and intelligent decision making. The popularity of terms such as Network Security, AI, Machine Learning, Information Management, DL, Embedded Systems, and Cyber-physical Systems further reflects DTAII research, which combines intelligent computing, secure communication, and advanced digital infrastructure. As seen in Figure 6, there are a lot of intertwined relationships between these keywords, which demonstrates the interdisciplinary nature of this research field. Table XI, the thematic clustering results, shows that there are 5 main research clusters in DTAII. The Red cluster (28 members) is dedicated to DT applications in the fields of information management, BIM, and intelligent infrastructure. The Green cluster (28 members) focuses on AI-related learning, optimization, and decision support. The Blue cluster (19 members) is connected to sustainability, smart cities, and infrastructure lifecycle management. The Yellow cluster (16 members) is mainly focused on Cybersecurity and Secure DT deployment, while the Purple cluster (12 members) is primarily on Blockchain-based architectures and decentralized DT frameworks. The KCN shows that DTAII research is moving towards intelligent, secure, and interoperable DT ecosystems, with the use of AI, IoT, BIM, cybersecurity, and advanced computing technologies.

2) Keyword Co-occurrence Network Analysis for SMSS: The KCN analysis for SMSS is illustrated in Figure 7, while the most influential keywords and thematic clusters are summarized in Tables XII and XIII, respectively. DT is the dominant keyword with the highest frequency (630) and link strength (2909), indicating its central role in the research area. As shown in Table XII, IoT (frequency 266, link strength 1675) exhibits the strongest association with DT, emphasizing its importance in real-time sensing and communication. Other significant keywords such as Decision Making, Learning System, Machine Learning, Information Management, AI, Industry 4.0, and Real-time System show how intelligent analytics and smart sensing technologies are also increasingly becoming part of DT-based monitoring systems. The high level of interconnections among these keywords in Figure 7 shows the multidisciplinary character of SMSS research. The thematic clustering results outlined in Table XIII show that five main research clusters exist in SMSS. The Red cluster (30 members) deals with IoT-integrated DT systems and secure infrastructure management. The Green cluster (28 members) focuses on information management and integration of BIM. The Blue cluster (23 members) relates to Industry 4.0, interoperability, and DT architectures. The Yellow cluster (19

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TABLE IX: Top prominent journals in each research area

Research Area	Journal Name	c	np	ACPP	np%	IF
DTAII	Applied Sciences Switzerland	505	47	10.74	7.33	2.5
	IEEE Access	475	44	10.80	7.41	3.6
	Sustainability Switzerland	433	33	13.12	5.19	3.3
	Sensors	389	32	12.16	3.67	4.5
	Advanced Engineering Informatics	238	21	11.34	3.28	9.9
SMSS	Applied Sciences Switzerland	4185	82	51.03	8.90	2.5
	IEEE Access	574	72	7.97	7.19	3.6
	Sensors	488	68	7.17	6.73	4.5
	Sustainability Switzerland	456	45	10.13	4.56	3.3
	IEEE Internet of Things Journal	326	43	7.58	4.26	8.9
ASID	IEEE Access	2129	54	39.42	4.41	3.6
	Applied Sciences Switzerland	1534	53	28.94	6.01	2.5
	Sensors	988	43	22.98	4.87	4.5
	Sustainability Switzerland	837	28	29.89	3.17	3.3
	Advanced Engineering Informatics	457	24	19.04	2.72	9.9
PIAM	Applied Sciences Switzerland	752	40	18.80	10.18	2.5
	IEEE Access	633	30	21.10	7.63	3.6
	Sensors	214	20	10.7	5.09	4.5
	Sustainability Switzerland	165	18	9.17	4.58	3.3
	Advanced Engineering Informatics	174	13	13.38	3.30	9.9

TABLE X: Top 10 keywords by frequency and link strength (DTAII)

Keyword	Frequency	Link Strength
Digital Twin	393	1939
Internet of Things	95	642
Decision Making	90	594
Network Security	62	469
Artificial Intelligence	62	361
Machine Learning	52	360
Information Management	58	354
Deep Learning	52	315
Embedded Systems	40	331
Cyber-physical System	33	308

members) highlights structural health monitoring and AI-based predictive analytics, whereas the Purple cluster (13 members) focuses on cyber-physical systems and digital transformation. In summary, the KCN suggests that the trend of intelligent DT infrastructure systems merging with IoT, AI, and Real-time monitoring has led research in SMSS.

3) Keyword Co-occurrence Network Analysis for ASID: The KCN analysis for ASID is shown in Fig. 8, and the most influential keywords and thematic clusters are presented in Tables XIV and XV, respectively. DT is the most frequently used keyword and has the strongest link (2538) and frequency (528), making it the focus word of the research field of structural intelligence supported by AI. As shown in Table XIV, AI (frequency 156, link strength 811), DL (frequency 153, link strength 854), Learning System (frequency 139, link strength 940), and Machine Learning (frequency 139, link strength 870) are the most strongly connected technologies, highlighting their importance in intelligent damage analysis and prediction. The use of intelligent sensing and DT technologies is also reflected in other prominent keywords, such as IoT, SHM, Information Management, and Embedded Systems. Thematic clustering in ASID identifies five main research clusters as indicated in Table XV below. The Red cluster (29 members) is devoted to predictive analytics and structural damage assessment. The Green cluster (24 members) focuses on AI-powered DT ecosystems and intelligent decision-making. The Blue cluster (19 members) is linked with cyber-physical systems and secure DT architectures. The Yellow cluster (14 members) emphasizes reinforcement learning and optimization methods, while the Purple cluster (11 members) is devoted to BIM, semantic modeling, and digital representation of built environments. There is a general trend to focus on the development and implementation of AI-powered predictive intelligence and next-generation DT applications in the field of SHM within the ASID research community, according to the KCN.

4) Keyword Co-occurrence Network Analysis for PIAM: The KCN analysis for PIAM is presented in Figure 9, and the most influential keywords and thematic clusters are presented in Tables XVI and XVII, respectively. The most important and high-frequency term is DT (268), which also has the biggest link strength (1201), giving it a central position in the prediction of infrastructure. Table XVI illustrates the strong association between DT and Predictive Maintenance (frequency 97, link strength 519), Decision Making (frequency 72, link strength 481), which are essential elements in the domain of intelligent asset management. The use of DT technologies for optimising the infrastructure lifecycle is evident in other prominent keywords such as Decision Support System, Asset Management, Information Management, IoT, Fault Detection, Building Information Modeling, and Interoperability. The thematic clustering results outlined in Table XVII reveal five clus-

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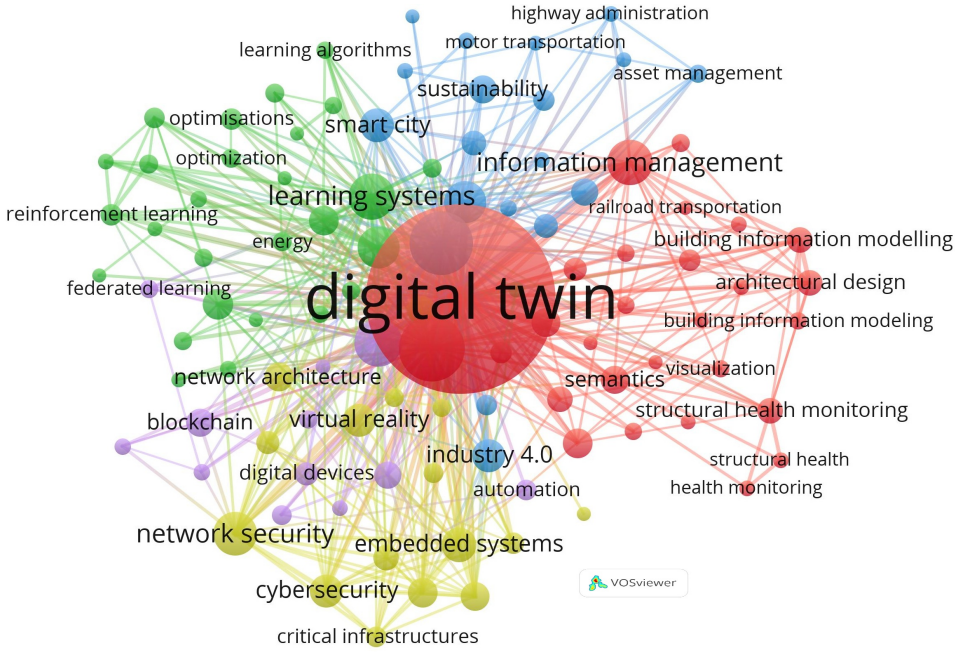


Fig. 6: Keyword co-occurrence network for DTAII

TABLE XI: Thematic clusters identified in DTAII

Cluster	Number of Members	Core Focus Areas
Red	28	Information management, Building Information Modeling (BIM), architectural design, semantics, visualization, and structural health monitoring for DT-enabled infrastructure and building systems.
Green	28	Learning systems, reinforcement learning, federated learning, optimization techniques, energy management, smart city applications, and AI-driven decision support for DT environments.
Blue	19	Sustainability, smart cities, transportation systems, asset management, highway administration, and infrastructure lifecycle management within DT ecosystems.
Yellow	16	Network security, cybersecurity, embedded systems, Industry 4.0, critical infrastructure protection, virtual reality, and secure DT deployment.
Purple	12	Blockchain-enabled DTs, network architectures, digital devices, automation frameworks, federated computing, and decentralized system integration.

TABLE XII: Top 10 keywords by frequency and link strength (SMSS)

Keyword	Frequency	Link Strength
Digital Twin	630	2909
Internet of Things	266	1675
Decision Making	110	726
Learning System	92	669
Machine Learning	86	616
Information Management	85	557
Artificial Intelligence	58	441
E-Learning	52	315
Industry 4.0	77	397
Real-time System	70	358

ters of major research within PIAM. The Red cluster (26 members) is dedicated to AI-powered DTs, interoperability, and decision support. The Green cluster (19 members) focuses on Industry 4.0, smart manufacturing, and lifecycle management. The Blue cluster (17 members) is related to fault diagnosis and predictive maintenance. The Yellow cluster (14 members) is dedicated to predictive analytics and optimization based on data, while the Purple cluster (14 members) is committed to structural health monitoring and real-time assessment of the state of the infrastructure. Overall, the KCN highlights that the field of PIAM research is increasingly focused on topics that involve intelligent maintenance, asset lifecycle optimization, and AI-driven DT technologies.

D. Country Collaboration Network Analysis

The CCN analysis is a common scientometric method to analyze international research collaboration and determine the top contributing countries within a research field. It visualizes co-authorship networks, allowing one to see the international knowledge-sharing system. In the current study, CiteSpace was used to construct the CCNs, where the node size represents the citation count of a country, with larger nodes indicating

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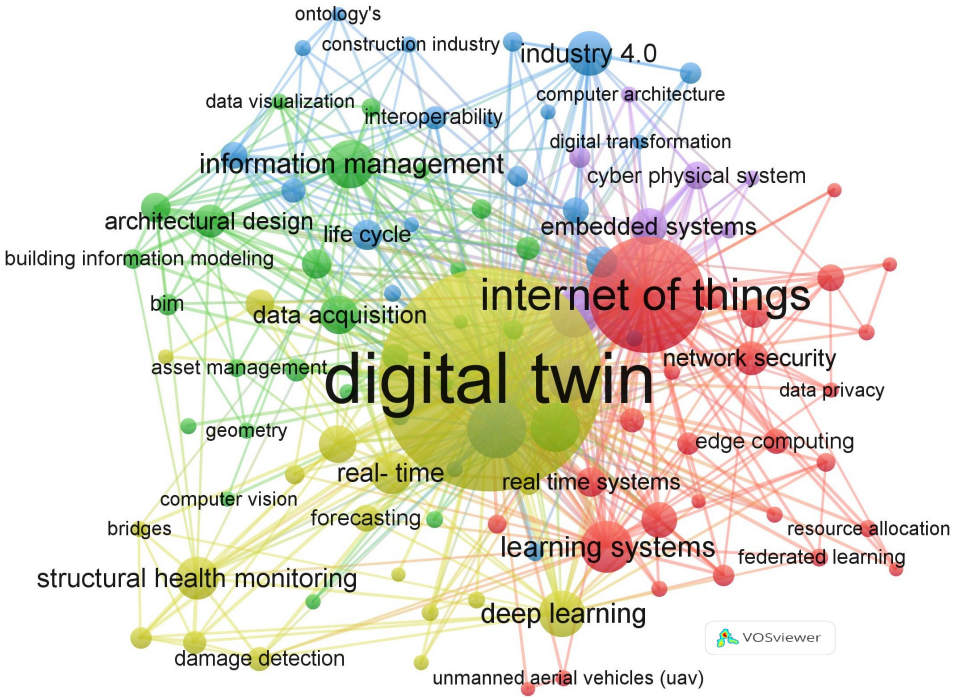


Fig. 7: Keyword co-occurrence network for SMSS

TABLE XIII: Thematic clusters identified in SMSS

Cluster	Number of Members	Core Focus Areas
Red	30	Internet of Things (IoT)-driven DT systems, network security, data privacy, edge computing, federated learning, real-time systems, and intelligent infrastructure management.
Green	28	Information management, architectural design, Building Information Modeling (BIM), asset management, data visualization, geometry modeling, and DT integration in the construction and built environment.
Blue	23	DT architectures, interoperability frameworks, Industry 4.0 technologies, lifecycle management, ontologies, construction industry applications, and system integration.
Yellow	19	Structural health monitoring, deep learning, damage detection, data acquisition, computer vision, forecasting, and AI-enabled predictive analytics for infrastructure monitoring.
Purple	13	Cyber-physical systems, embedded systems, digital transformation, intelligent device connectivity, and real-time interaction between physical and virtual environments.

greater scientific influence. The purple (pink) ring around a node shows betweenness centrality, with thicker rings meaning countries play a significant role as a bridge between different research communities. The edges connecting nodes represent collaboration relationships, and thicker links indicate more intense collaboration among countries. Nations with more nodes, higher centrality, and stronger collaborative ties tend to be more influential in driving global research progress and enabling knowledge sharing.

1) Country Collaboration Network Analysis for DTAII : The CCN for DTAII is illustrated in Figure 10, while the leading collaborative countries are summarized in Table XVIII. The generated network contains 84 nodes and 338 links of collaboration with a network density of 0.097. The network construction was based on a g-index of 25, and the silhouette value of 0.89 suggests a stable and clear collaboration structure. As indicated in Table XVIII, China (135 citations) is the top country, followed by the United Kingdom (84 citations), the United States (81 citations), Italy (57 citations), and Spain (53 citations). The United Kingdom has the highest degree (40) and centrality (0.35), which is highly indicative of the importance of the UK in establishing links between research collaborations throughout the world. The United States also has good collaborative relationships, with a degree of 33 and centrality of 0.23; Italy and Spain also play an important role as regional collaborators. In general, the CCN shows that research productivity is high and that international collaboration is well established, with China being the main source of research and the UK being the main source of international scientific collaboration.

2) Country Collaboration Network Analysis for SMSS: The CCN for SMSS is shown in Figure 11, while the summary of top countries in collaboration is shown in Table XIX in terms of number of citations, degrees, and centrality. The

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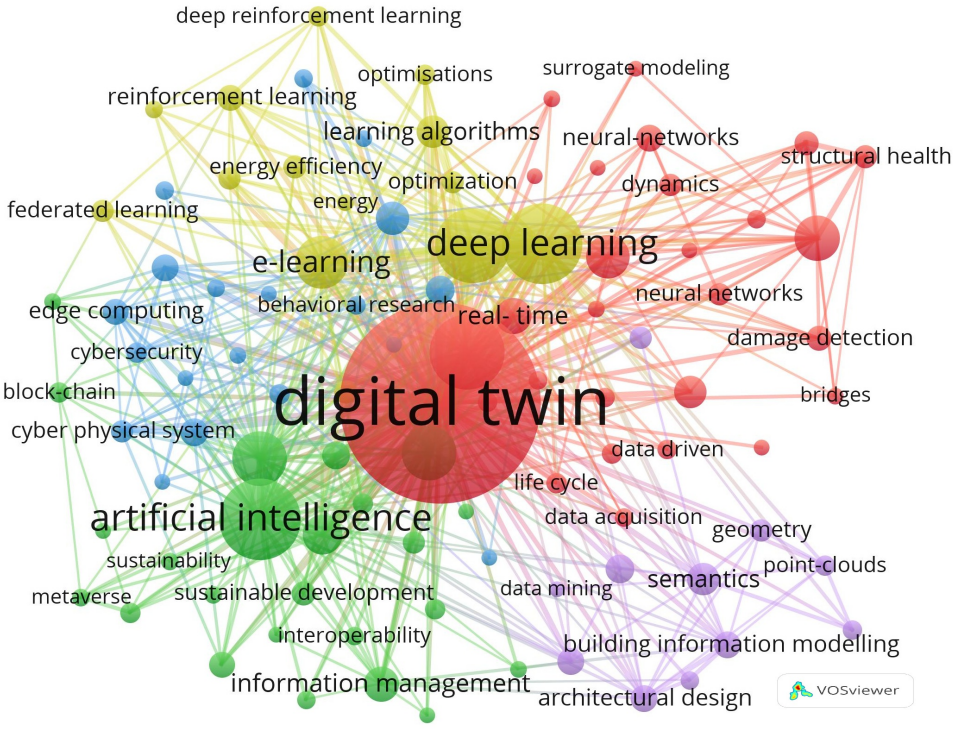


Fig. 8: Keyword co-occurrence network for ASID

TABLE XIV: Top 10 keywords by frequency and link strength (ASID)

Keyword	Frequency	Link Strength
Digital Twin	528	2538
Learning System	139	940
Machine Learning	139	870
Deep Learning	153	854
Artificial Intelligence	156	811
Internet of Things	92	664
E-learning	88	559
Structural Health Monitoring	72	346
Information Management	50	321
Embedded System	45	304

collaboration network has 82 nodes, 349 links, and a network density of 0.105. The network is generated with a g-index of 25, which gives a silhouette value of 0.89, representing a good quality of network collaboration. As shown in the results, China has the highest research impact with 271 citation counts, indicating that China contributes a significant amount of research in SMSS. By contrast, the UK has the highest degree (41) and centrality (0.33), reflecting its importance as the central link between the international research communities. The United States is at the top with 122 citations, followed by Italy (92) and Germany (63), leading in the global network of collaborative activity. In aggregate, the CCN provides a picture of a highly connected international research environment where China has a very strong scientific output and where the United Kingdom and the United States are key in building global research collaboration and knowledge exchange.

3) Country Collaboration Network Analysis for ASID: The international collaboration pattern for ASID is presented in Figure 12, and the collaboration metrics are summarized in Table XX. It creates a network of 76 nodes and 170 collaboration links, and the network density is 0.059. The network, using the g-index set to 25, has a silhouette of 0.94, which reflects a structure of collaboration that is highly coherent. China has the largest number of citations (254) in the collaboration landscape, showing that it has made a strong research contribution. The United States is the main collaboration centre with the highest degree (43) and centrality (0.35), showing its high international connectivity. Collaborative activity in the United Kingdom is also very high (106 citations), and India and Italy also contribute to the global research network by being active in research. In general, the CCN reflects a well-balanced, international collaborative working environment, with China leading research productivity and the United States playing a significant role in facilitating knowledge sharing and building collaborative partnerships in ASID research.

4) Country Collaboration Network Analysis for PIAM: The collaborative nature of PIAM is illustrated in Figure 13 and the country-level metrics summarized in Table XXI. The collaboration network consists of 87 nodes and has a network density of 0.085 and is generated based on a g-index of 25. The silhouette score is 0.67, which is a good level

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TABLE XV: Thematic clusters identified in ASID

Cluster	Number of Members	Core Focus Areas
Red	29	DT-driven predictive analytics, deep learning, neural networks, structural health monitoring, damage detection, real-time systems, lifecycle management, surrogate modeling, and data-driven infrastructure intelligence.
Green	24	Artificial intelligence, information management, interoperability, sustainable development, metaverse technologies, sustainability assessment, and intelligent decision-making for DT ecosystems.
Blue	19	Cyber-physical systems, edge computing, cybersecurity, blockchain, behavioral research, data integration, and secure DT architectures for distributed environments.
Yellow	14	Deep reinforcement learning, reinforcement learning, learning algorithms, optimization techniques, energy efficiency, federated learning, e-learning, and AI-based optimization frameworks.
Purple	11	Building Information Modeling (BIM), architectural design, semantics, geometry, point clouds, data mining, and digital representation of built environments for DT applications.

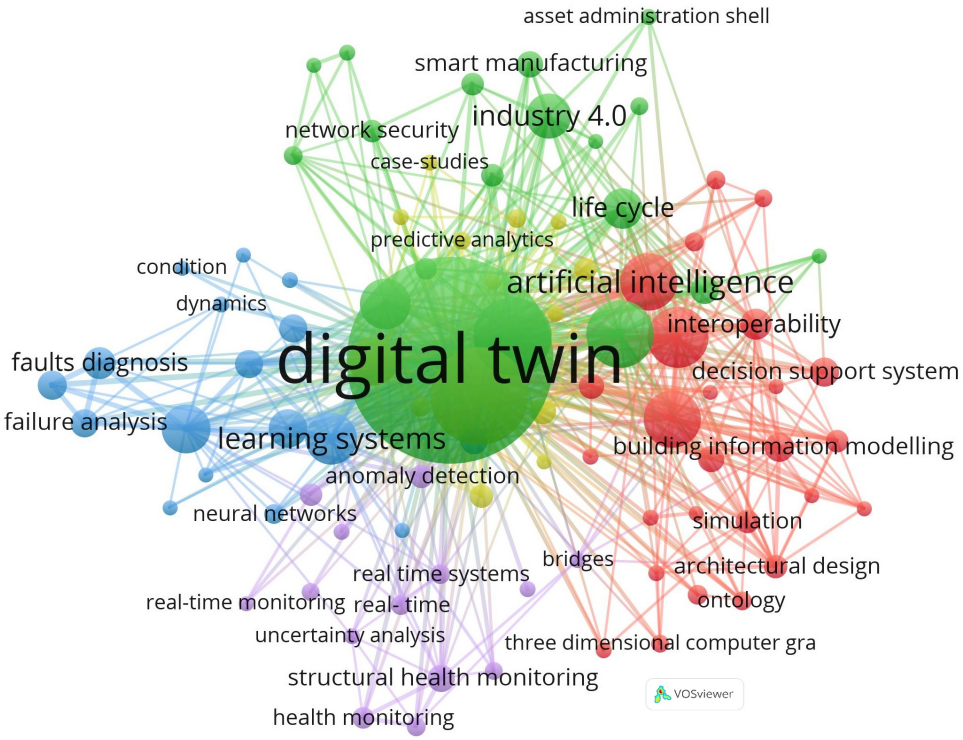


Fig. 9: Keyword co-occurrence network for PIAM

of coherence in the collaboration structure. China is the top contributor with 86 citations, indicating high research activity in PIAM. The United Kingdom has the highest degree (22) and centrality (0.26) of collaborative influence, indicating its active role in the network of international research partners. Notable collaborative participation also exists between Italy and the United States, with Germany participating in the enhancement of the overall research network. In general, CCN concludes that there is a well-founded international collaboration framework found in PIAM research activities, led by China, and with an important role played by European countries in supporting the cross-national scientific cooperation activities.

E. Document Co-citation Analysis

The DCA is one of the most popular scientometric methods for determining the intellectual structure and significant publications of a research field. It analyzes the co-citation frequency of two documents, which reflects the knowledge relationship between the two documents with a higher co-citation frequency. In the current study, CiteSpace was used to construct the DCA networks, where the node size represents the citation frequency of a document, the purple (pink) ring indicates betweenness centrality, the links denote co-citation relationships, and the link thickness reflects the strength of co-citation. Larger nodes and higher centrality generally mean more influential studies that have influenced the field of research.

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TABLE XVI: Top 10 keywords by frequency and link strength (PIAM)

Keyword	Frequency	Link Strength
Digital Twin	268	1201
Predictive Maintenance	97	519
Decision Making	72	481
Decision Support System	60	336
Information Management	48	297
Asset Management	52	296
Internet of Things	41	253
Fault Detection	39	225
Building Information Modeling	20	160
Interoperability	20	160

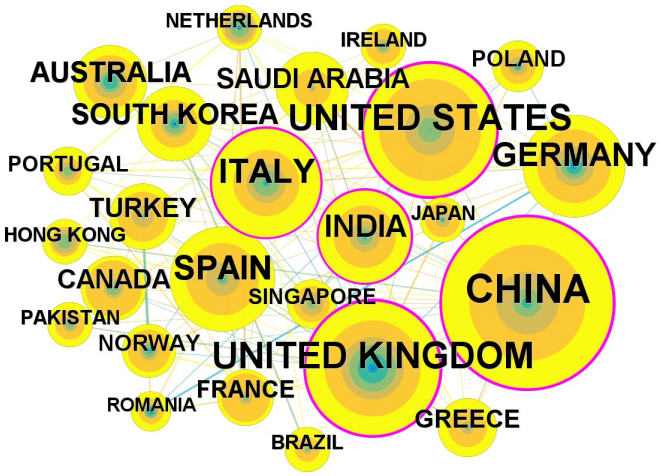


Fig. 10: Country collaboration network for DTAII

1) Document Co-citation Analysis for DTAII : The DCA for DTAII is shown in Figure 14, and the main co-citation clusters are summarised in Table XXII. The generated network consists of 389 nodes and 1029 links and has a density of 0.0136. The density is relatively low, which is common for co-citation networks, but still indicates that the knowledge base is well structured; it is not an indication of weak co-citation. The quality of the network is also validated with a Modularity value of 0.77, which shows that the research themes are well-distinguished from each other, and a Mean Silhouette value of 0.89, which indicates that the research themes in the clusters are reliable. Based on the g-index value of 25, five major clusters were found, which correspond to the major knowledge domains of the research of DTAII.

1) **Cluster #1: Advanced Rail Transit System:** It is the largest cluster with 45 publications and a Silhouette value of 0.819. The cluster highlighted the emerging opportunities of DT technologies for intelligent railway systems and real-time monitoring of infrastructure and smart transportation management. The work by Crespi N et al. [51] is the most influential reference in this cluster, receiving 9 co-citations.

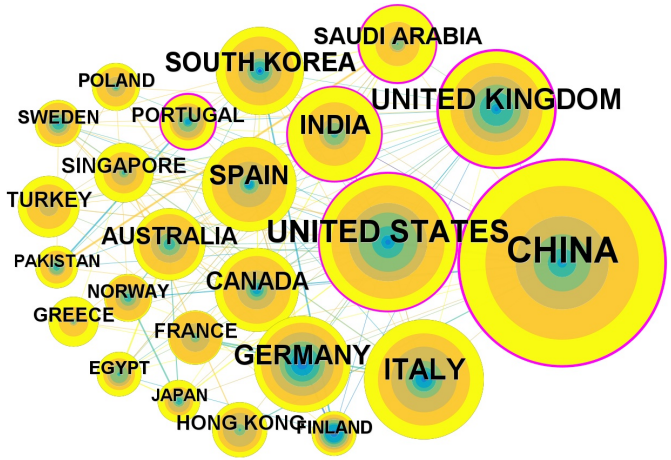


Fig. 11: Country collaboration network for SMSS

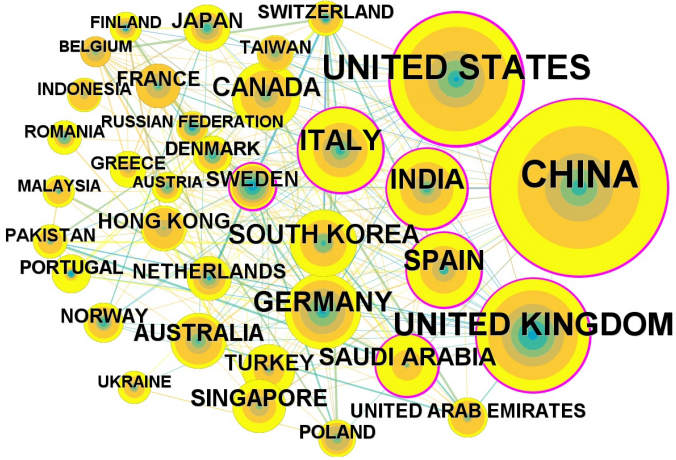


Fig. 12: Country collaboration network for ASID

2) **Cluster #2: Digital Twin Technology:** It is composed of 37 publications having a Silhouette value of 0.891. It comprises the fundamental technological advancements in DTs, such as system architectures, integration frameworks, and enabling technologies. Hicks B et al. [52] is the paper that is cited most often, with 8 citations.
3) **Cluster #4: Road Infrastructure Management:** It has 32 publications with a Silhouette value of 0.910. The research focus extends predominantly to intelligent road maintenance, asset management, and road transportation infrastructure, where DT technologies play an important role. Gangolells M et al. [53] is the most influential in this cluster, with 9 co-citations.
4) **Cluster #5: Sustainable Urban Governance Network:** It includes 28 publications with a Silhouette value of 0.899. The research in this cluster is not limited to infrastructure monitoring but is aimed at a wider range of urban governance, emphasising the potential of

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TABLE XVII: Thematic clusters identified in ASID

Cluster	Number of Members	Core Focus Areas
Red	26	Artificial intelligence-driven DTs, interoperability, decision support systems, Building Information Modeling (BIM), simulation, architectural design, ontology development, and intelligent infrastructure management.
Green	19	Industry 4.0, smart manufacturing, lifecycle management, predictive analytics, asset administration shell, network security, and AI-enabled industrial DT applications.
Blue	17	Fault diagnosis, failure analysis, condition monitoring, system dynamics, learning systems, neural networks, anomaly detection, and predictive maintenance in DT environments.
Yellow	14	Predictive analytics, anomaly detection, machine learning applications, case studies, optimization techniques, and data-driven decision-making for DT systems.
Purple	14	Structural health monitoring, health monitoring, real-time monitoring, uncertainty analysis, real-time systems, bridge monitoring, and infrastructure performance assessment.

TABLE XVIII: Top 5 countries by citation count, degree, and centrality (DTAII)

Rank	Country	Citation Counts	Degree	Centrality
1	China	135	32	0.24
2	United Kingdom	84	40	0.35
3	United States	81	33	0.23
4	Italy	57	27	0.11
5	Spain	53	23	0.08

TABLE XIX: Top 5 countries by citation count, degree, and centrality (SMSS)

Country	Citation Counts	Degree	Centrality
China	271	33	0.27
United States	122	32	0.17
Italy	92	22	0.12
United Kingdom	89	41	0.33
Germany	63	25	0.07

TABLE XX: Top 5 countries by citation count, degree, and centrality (ASID)

Rank	Country	Citation Counts	Degree	Centrality
1	China	254	34	0.20
2	United States	144	43	0.35
3	United Kingdom	106	34	0.17
4	Italy	61	23	0.11
5	India	54	25	0.14

TABLE XXI: Top 5 countries by citation count, degree, and centrality (PIAM)

Rank	Country	Citation Counts	Degree	Centrality
1	China	86	17	0.28
2	United Kingdom	42	22	0.26
3	Italy	40	13	0.17
4	United States	33	15	0.18
5	Germany	27	13	0.13

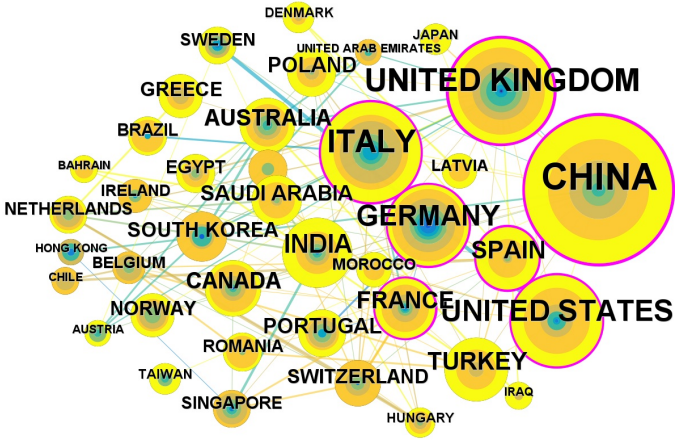


Fig. 13: Country collaboration network for PIAM

DTs to contribute to smart cities and sustainable urban development. The paper by Barlow C et al. [37] (21 citations) is the main reference for this line of research.

5) **Cluster #6: Blockchain-Based Digital Twin:** It is a cluster of 25 publications with a Silhouette value of 0.907. The cluster focuses on the advancements made in recent years that have enabled the development of smart contracts and the application of blockchain smart contracts to create DT systems for enhancing security, trust, and decentralised data management. Fogli D et al. [54] have received 13 co-citations and is the most influential publication.

As a whole, the co-citation network indicates that the scope of DTAII research has been gradually expanded beyond transportation-related DT applications to a wider range of topics such as smart urban governance, secure DT architectures, and emerging decentralized technologies.

2) Document Co-citation Analysis for SMSS: The DCA for SMSS is shown in Fig. 15 and consists of 462 nodes and 1298 links with a network density of 0.0122. The relatively low density of the network does not detract from the fact that it reflects a clear intellectual structure of the research field. The clustering results are high quality as the Modularity value is 0.7642 and the Mean Silhouette value is 0.8942, which means that the clustering results are separated and consistent. Six

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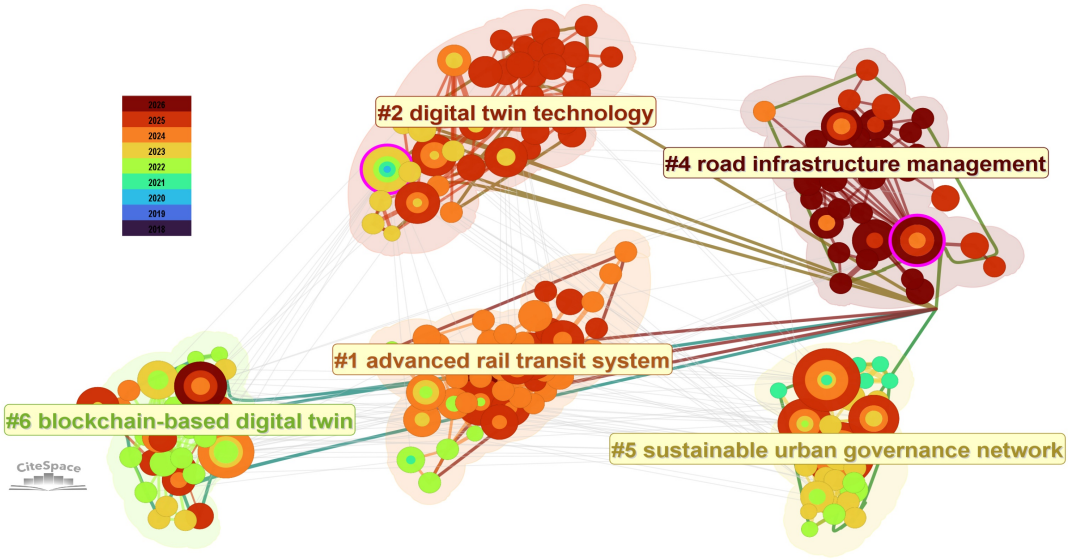


Fig. 14: Document co-citation network for DTAII

TABLE XXII: Top five document co-citation clusters and their most influential publications (DTAII)

Cluster ID	Size	Silhouette	Label (LLR)	Average Year	Most Cited Article	Citation Count
#1	45	0.819	Advanced Rail Transit System	2020	Crespi N et al. [51]	9
#2	37	0.891	Digital Twin Technology	2021	Hicks B et al. [52]	8
#4	32	0.910	Road Infrastructure Management	2022	Gangolells M et al. [53]	9
#5	28	0.899	Sustainable Urban Governance Network	2019	Barlow C et al. [37]	21
#6	25	0.907	Blockchain-Based Digital Twin	2019	Fogli D et al. [54]	13

major clusters were created based on a g-index of 25, which is equivalent to the major themes of SMSS. The summary of cluster characteristics is given in Table XXIII.

1) **Cluster #1: Multifidelity Surrogate Model:** It is the largest cluster, consisting of 48 publications (Silhouette value = 0.917). The cluster focuses on surrogate modeling, computational simulation, and DT-based prediction methods for smart monitoring systems. The most influential publication is the publication by Kvamsdal T et al. [34] (32 citations).
2) **Cluster #2: Framework-Enabled Asset Lifecycle Management:** This cluster contains 46 publications with a Silhouette value of 0.878. It highlights DT models, asset lifecycle management, infrastructure monitoring, and optimization of maintenance. The most co-cited publication, with 34 co-citations, is the one by Barlow C et al. [37].
3) **Cluster #3: Smart Cities:** It (43 publications, Silhouette value 0.879) focuses on the use of DTs in the development of smart cities, intelligent urban services, and management of digital infrastructure in cities. Clarke S et al. [55] is the most often cited paper with 14 citations.
4) **Cluster #4: Building Energy Consumption Forecasting:** It contains 32 publications that have a Silhouette value of 0.820. The research work mainly focuses on energy forecasting, intelligent building management, and sustainability via DT technologies. The publication by Crespi N et al. [51] is the most influential, with 25 citations.
5) **Cluster #5: Healthcare Revolution:** With the highest Silhouette value of 0.922, this cluster has excellent cluster consistency. It discusses the use of DTs in healthcare systems, patient monitoring, and intelligent medical services. The most influential publication is by Zhang M et al. [56], with 10 citations.
6) **Cluster #6: Large Underground Cavern:** This cluster also contains 29 publications with a Silhouette value of 0.919. The research centers on underground structures, geotechnical engineering, and the monitoring of large underground structures using DTs. The most influential publication is the one by Devine D et al. [57], which is cited 12 times.

In summary, the DCA shows that SMSS research is not limited to the monitoring of infrastructure, but includes asset lifecycle management, smart cities, intelligent buildings, the healthcare industry, and underground engineering. The clusters were used to show the growing potential of the DT technologies for the support of intelligent monitoring and decision-making in different application fields.

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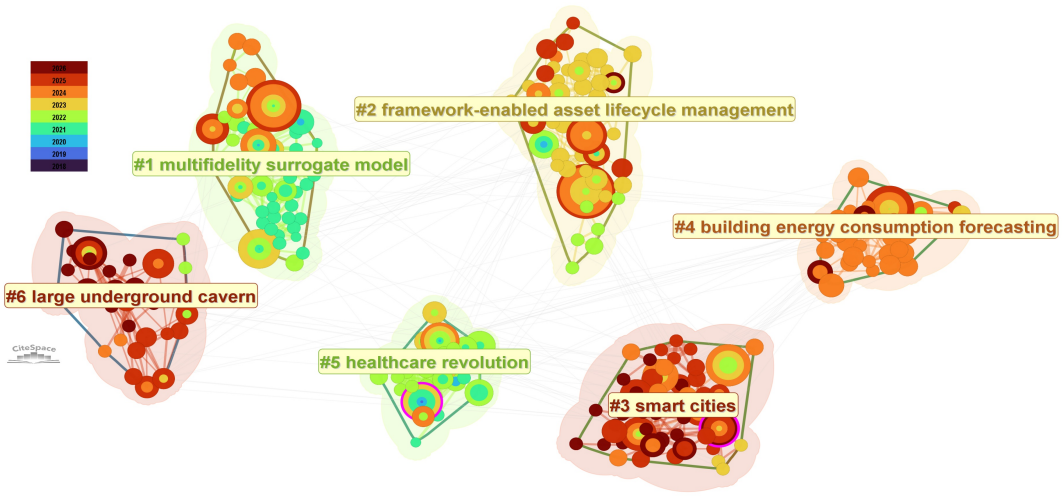


Fig. 15: Document co-citation network for SMSS

TABLE XXIII: Major document co-citation clusters and their most influential publications (SMSS)

Cluster ID	Size	Silhouette	LLR Label	Average Year	Most Cited Article	Citation Count
#1	48	0.917	Multifidelity Surrogate Model	2018	Kvamsdal T et al. [34]	32
#2	46	0.878	Framework-Enabled Asset Lifecycle Management	2020	Barlow C et al. [37]	34
#3	43	0.879	Smart Cities	2022	Clarke S et al. [55]	14
#4	32	0.820	Building Energy Consumption Forecasting	2021	Crespi N et al. [51]	25
#5	29	0.922	Healthcare Revolution	2018	Zhang M et al. [56]	10
#6	29	0.919	Large Underground Cavern	2022	Devine D et al. [57]	12

3) Document Co-citation Analysis for ASID: The DCA for ASID, shown in Figure 16, consists of 433 nodes and 1063 links, resulting in a network density of 0.0114. The network is relatively low density, but it reflects the intellectual architecture of the research field to a good degree. The clustering quality is indicated by a Modularity of 0.8471, which shows that the research themes are well separated from each other, and a Mean Silhouette value of 0.9448, which shows that the identified clusters are reliable and well formed. Six major clusters were found based on a g-index of 25, which are the main areas of research and development in ASID. The details of these clusters are summarized in Table XXIV.

1) **Cluster #0: The-Art Review:** It is the largest cluster, comprising 40 publications with a Silhouette value of 0.920. The cluster mainly represents review studies that summarize recent developments in DT technologies, artificial intelligence, and intelligent structural analytics. The most cited publication is the one by Barlow C et al. [37], with 24 citations.
2) **Cluster #2: Comprehensive Survey:** It contains 33 publications with a Silhouette value of 0.900. It highlights survey papers covering DT frameworks, implementation methods, and recent research areas. The most influential paper is the one by kritzinger et al. [58] with 14 citations.
3) **Cluster #4: PV System:** Comprising 27 publications with a Silhouette value of 0.928, this cluster highlights DT applications in photovoltaic systems, renewable energy optimization, and intelligent energy management. The publication by Karimi M et al. [59] had 10 citations, which made it the most cited paper.
4) **Cluster #5: Digital Twin Modeling:** It contains 25 publications with a Silhouette value of 0.962, which is very good consistency. The research focuses mainly on DT modelling, simulation approaches, and predictive intelligence of engineering systems. The most influential work, regarding its number of co-citations, is published by Li Z et al. [60] (29 citations).
5) **Cluster #7: Industrial Facilities:** It includes 21 publications with a Silhouette value of 0.962. It is oriented towards applications related to the DT of industrial facilities, smart manufacturing, and infrastructure monitoring. Chen K et al. [61] publication is the most frequently co-cited one, with 6 co-citations.
6) **Cluster #9: Open Issue:** It is the smallest cluster, with 18 publications having the highest Silhouette value of 0.967. It identifies existing research challenges, limitations, and potential future opportunities for AI-enabled DT systems. The publication by Newman ST et al. [62] is the most influential reference, receiving 10 citations.

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Overall, the DCA reveals that ASID research has evolved from comprehensive review studies toward specialized applications involving DT modeling, renewable energy systems, industrial facilities, and emerging research challenges. These clusters collectively illustrate the expanding knowledge base and research maturity of AI-enabled structural intelligence and damage analytics.

4) Document Co-citation Analysis for PIAM: The DCA for PIAM, illustrated in Figure 17, consists of 320 nodes and 782 links, resulting in a network density of 0.0153. The network has a clear knowledge structure with a Modularity value of 0.8313, which shows a clear separation between the identified research themes. Moreover, the Mean Silhouette value of 0.9448 shows that the created clusters are consistent and reliable. A g-index value of 25 resulted in the identification of six large clusters, which correspond to the main research directions of predictive infrastructure intelligence and asset management. The characteristics of these clusters are presented in Table XXV.

1) **Cluster #0: Digital Twin-Inspired Method:** It represents the biggest cluster containing 30 publications and has a Silhouette value of 0.964. The cluster emphasizes DT-related approaches towards intelligent asset management, predictive maintenance, and infrastructure decision-making. Lakhssassi A et al. [63] is the most influential article, with 7 citations.
2) **Cluster #1: Blockchain-Empowered Digital Twins Collaboration:** This cluster consists of 29 publications with a Silhouette value of 0.917. Research focuses on collaboration in the DT, secure information sharing, and decentralized asset management with blockchain technology. The most frequently co-cited work is the publication of Gardner et al. [64], which had 15 citations.
3) **Cluster #2: Additive Manufacturing:** This cluster (25 publications, Silhouette value 0.969) examines the use of DT in the context of additive manufacturing, intelligent production, and smart industrial processes. The Tao et al. [65] is the most cited study, with 12 citations.
4) **Cluster #3: Novel Ontology:** It consists of 23 publications with a Silhouette value of 0.936. The research primarily focuses on the representation of knowledge, semantic interoperability, and information integration in the context of DT systems. The most influential publication is the work by Holmstrom J et al. [66], with 6 citations.
5) **Cluster #4: Blockchain-Based Digital Twin:** There are 22 publications that make up the cluster with a Silhouette value of 0.933. It emphasizes DT frameworks supported by blockchain to manage data securely, track it, and provide trusted infrastructure services. The work by Zhang M et al. [56] was the most cited (9 citations).
6) **Cluster #5: Intelligent Maintenance System:** It consists of 21 publications with a Silhouette value of 0.935. Research is focused on intelligent maintenance strategies, condition monitoring, and predictive asset management using the DT approach. Tao et al. [67] is cited the most, with 2 citations.

Overall, the DCA demonstrates that PIAM research has evolved from intelligent maintenance and DT methodologies toward blockchain-enabled collaboration, semantic interoperability, and advanced manufacturing applications. These clusters collectively illustrate the growing integration of intelligent technologies for predictive infrastructure management and asset lifecycle optimization.

V. Discussion and Result

The findings show that AI-integrated DT and SHM have become a fast-growing multidisciplinary research field with notable progress in smart sensing, data analysis-based structural assessment, and predictive infrastructure management. The integrated scientometric analyses give a thorough picture of thematic evolution, research productivity, international cooperation, and the intellectual structure of the field, which helps answer the research questions. Four main research areas were identified by BERTopic, namely DTAII, SMSS, ASIDA, and PIAM, which correspond to different phases of the development of intelligent SHM. DTAII mainly targets DT architectures, IoT, Cyber-Physical Systems, and the implementation of intelligent integration of infrastructure, while SMSS concentrates on real-time sensing via IoT, wireless sensor networks, computer vision, UAV-assisted inspection, multi-sensor fusion, and Edge AI-based monitoring. ASIDA has moved beyond traditional AI, ML, and DL methods to explainable AI (XAI), PINNs, computer vision damage detection and structural anomaly detection, and is increasingly focused on trustworthy, interpretable, and physics-aware structural intelligence. Likewise, PIAM has progressed from predictive maintenance to RUL prediction, infrastructure resilience assessment, risk-aware maintenance optimisation, autonomous asset management, and AI-supported decision support systems, these aimed at supporting proactive maintenance for critical infrastructure throughout its lifecycle. Publication and citation analyses reveal a sharp increase in research activity after 2021, with SMSS representing the most mature research domain and ASIDA exhibiting the fastest growth, highlighting the increasing importance of intelligent analytics in SHM. China is still the main contributor in terms of research productivity, while the United States, the United Kingdom, Italy, Germany, and India are the main international collaboration hubs. Moreover, IEEE journals such as IEEE Access, Sensors, Applied Sciences, and Advanced Engineering Informatics have significantly helped to spread the research in this field. The KCN, CCN, and DCA analyses further demonstrate that the intellectual focus of the field is shifting from foundational DT architectures toward edge-cloud DTs, blockchain-enabled secure DT frameworks, federated DT ecosystems, intelligent sensing platforms, explainable and physics-informed AI, and autonomous predictive maintenance. Such trends are a sign of shifting from individual structural monitoring solutions to secure, explainable, autonomous, and resilient DT ecosystems that will enable real-time management of infrastructure and support long-term decision-making.

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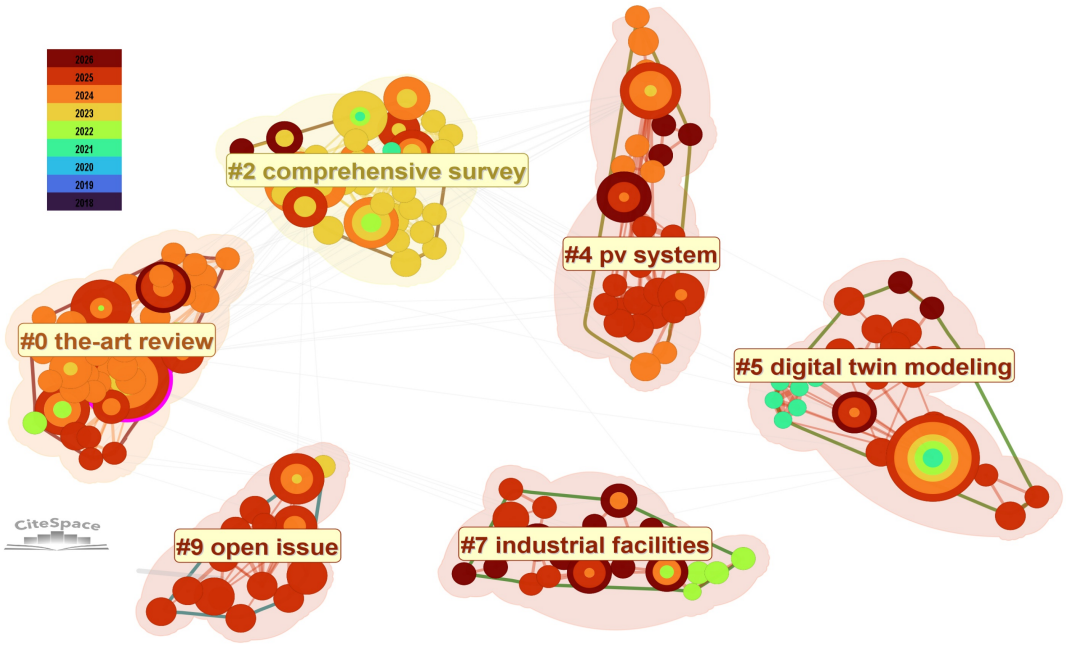


Fig. 16: Document co-citation network for ASID

TABLE XXIV: Major document co-citation clusters and their most influential publications (ASID)

Cluster ID	Size	Silhouette	LLR Label	Average Year	Most Cited Article	Citation Count
#0	40	0.920	The-Art Review	2020	Barlow C et al. [37]	24
#2	33	0.900	Comprehensive Survey	2020	kritzinger et al. [58]	14
#4	27	0.928	PV System	2022	Karimi M et al. [59]	10
#5	25	0.962	Digital Twin Modeling	2021	Li Z et al. [60]	29
#7	21	0.962	Industrial Facilities	2022	Chen K et al. [61]	6
#9	18	0.967	Open Issue	2021	Newman ST et al. [62]	10

TABLE XXV: Major document co-citation clusters and their most influential publications (PIAM)

Cluster ID	Size	Silhouette	LLR Label	Average Year	Most Cited Article	Citation Count
#0	30	0.964	Digital Twin-Inspired Method	2022	Lakhssassi A et al. [63]	7
#1	29	0.917	Blockchain-Empowered Digital Twins Collaboration	2019	Gardner et al. [64]	15
#2	25	0.969	Additive Manufacturing	2020	Tao et al. [65]	12
#3	23	0.936	Novel Ontology	2020	Holmstrom J et al. [66]	6
#4	22	0.933	Blockchain-Based Digital Twin	2018	Zhang M et al. [56]	9
#5	21	0.935	Intelligent Maintenance System	2017	Tao et al. [67]	2

A. Challenges and Potential Solutions

Table XXVI summarizes the major challenges and potential solutions across the four research domains. While each domain addresses distinct aspects of AI-integrated DT-enabled SHM, common challenges include data interoperability, scalability, real-time processing, and secure decision-making. The suggested solutions focus on standardisation of DT frameworks, AI-based analytics, edge-cloud computing, blockchain-based security, and intelligent asset management. The results underscore the importance of integrated, scalable, and trusted DT ecosystems for next-generation Structural Health Monitoring applications.

B. Research Limitations

Despite providing a comprehensive scientometric analysis, the current study has some limitations.

1) First, the analysis is based solely on the Scopus database and publications from 2018-2026. For the future, more databases such as Web of Science, IEEE Xplore, ACM Digital Library, etc., can be included to

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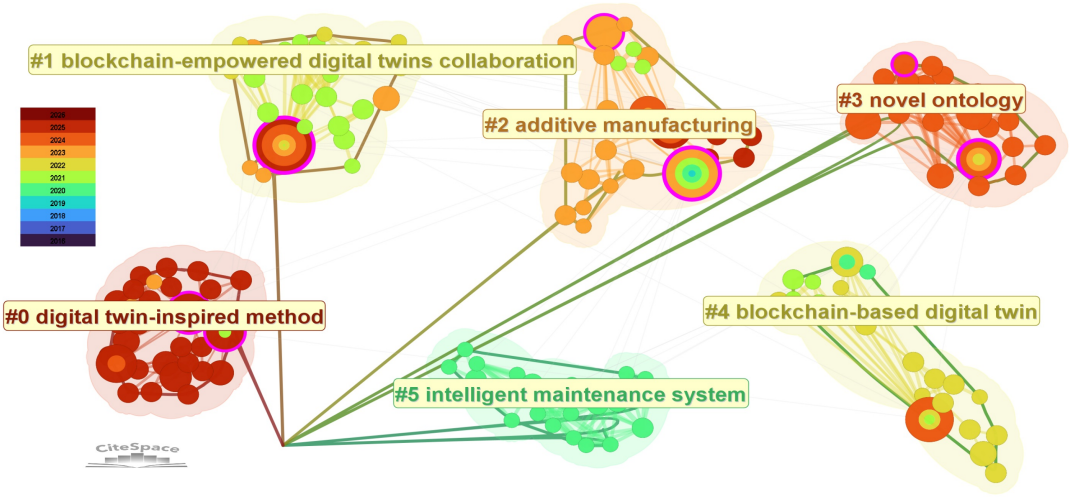


Fig. 17: Document co-citation Analysis for PIAM

TABLE XXVI: Challenges and potential solutions across each research domain

Category	Challenges	Potential Solutions
DTAII [14][68][69]	• Heterogeneous data integration and interoperability. • Real-time physical-virtual synchronization. • Cybersecurity and data privacy. • Lack of standardized DT architectures.	• Develop standardized DT frameworks. • Adopt edge-cloud collaborative computing. • Integrate blockchain for secure data sharing. • Employ semantic interoperability standards.
SMSS [70][71]	• Sensor reliability and communication latency. • Noisy and incomplete sensing data. • Energy consumption of sensor networks. • Scalability of real-time monitoring.	• AI-based sensor fusion. • Edge intelligence with adaptive sensing. • Energy-efficient IoT and 5G/6G networks. • Self-calibrating sensing frameworks.
ASIDA [72] [73]	• Limited labeled datasets. • Poor interpretability of AI models. • Weak model generalization. • High computational complexity.	• Explainable Artificial Intelligence (XAI). • Transfer and self-supervised learning. • Physics-informed AI models. • Federated learning and multimodal fusion.
PIAM [74][75]	• Uncertainty in Remaining Useful Life prediction. • Limited maintenance integration. • Decision uncertainty. • Lifecycle optimization.	• Hybrid DT–AI prognostic models. • Probabilistic prediction methods. • Reinforcement learning-based maintenance. • DT-enabled intelligent asset management.

enhance the coverage and decrease the bias coming from the specific database.

2) Second, the current study considered only English-language publications. It made it easier to collect the data and analyze it, but it also meant that studies written in other languages might have been missed. Future research may include multilingual literature in order to obtain a wider and more complete picture of the research landscape.
3) Finally, thematic and scientometric analysis is dependent on BERTopic, KCN, CCN, and DCA. The use of other techniques, such as bibliographic coupling, author co-citation, institutional collaboration, and citation burst analysis, can help to identify the intellectual structure and evolution of the field.

C. Future Research Directions

Based on the findings of the current scientometric analysis, several promising research directions are identified for advancing AI-integrated DTs in SHM.

1) **Explainable and Physics-informed AI:** The integration of XAI and PINNs with DT frameworks is another important direction for future research to enhance the interpretability, reliability, and trustworthy decision-making processes for critical infrastructure systems.
2) **Edge-enabled and Federated Digital Twins:** Edge-cloud and federated DT architectures can facilitate scalable, low-latency, and privacy-preserving DT structural

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monitoring for geographically distributed infrastructure systems.

3) **Foundation Models and Multi-modal Data Fusion:** The potential of emerging foundation models, computer vision, UAV-based inspection, and multi-modal sensor fusion for improved automated damage detection, condition assessment, and predictive analytics.
4) **Autonomous Predictive Maintenance:** Self-maintenance of infrastructure assets with the use of AI algorithms requires further research in the field of maintenance optimization, RUL prediction, reinforcement learning, and intelligent decision support systems.
5) **Secure and Interoperable DT Ecosystems:** Research is needed to develop standardized, interoperable, and blockchain-enabled DT frameworks that ensure secure data sharing, cyber resilience, and seamless integration across heterogeneous sensing platforms.
6) **Real-world Validation and Benchmarking:** Future research should be validated with large-scale real-world case studies, openly available benchmark datasets, and cross-domain infrastructure applications for practical deployment and technology transfer of emerging DT technologies.

VI. Conclusion

AI-based DTs are driving advances in SHM by enabling intelligent sensing, real-time structural assessment, and predictive infrastructure management. A comprehensive scientometric analysis of 2,930 Scopus-indexed publications from 2018 to 2026 was conducted using BERTopic, KCN, CCN, and DCA to examine the development, intellectual structure, collaboration, and emerging trends of the field. Four major research domains were identified: DTAII, SMSS, ASIDA, and PIAM, providing a structured view of the evolution of AI-enabled DT research for SHM. Research activity increased significantly after 2021, with ASIDA showing the fastest growth and SMSS emerging as the most mature domain. The analysis further indicates a transition from conventional DT architectures toward XAI, PINNs, Edge AI, federated DT ecosystems, foundation models, intelligent sensing, and autonomous predictive maintenance. The proposed taxonomy, scientometric findings, and research roadmap provide a foundation for advancing scalable, secure, explainable, and intelligent DT ecosystems for next-generation SHM and smart infrastructure.

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